CPR Esoft Quiz-6

Question #: 1

A 60-year-old woman comes to the physician because of a 2-month history of progressive weakness, loss of appetite and nausea. Her temperature is 36.5°C (97.7°C), pulse is 84/min, respirations are 20/min, and blood pressure is 160/90 mm Hg. Physical examination shows marked paleness and pitting pedal edema. Her 24-hour urine volume is 576 mL, urine creatinine concentration is 230 mg/dL, and plasma creatinine is 8 mg/dL. Which of the following values best approximates her creatinine clearance?

A. 3 mL/min
B. 6 mL/min
C. 9.5 mL/min
✓D. 11.5 mL/min
E. 15 mL/min

Rationale: Creatinine clearance = Ucr x V / Pcr
Ucr = urine creatinine concentration = 230.
Pcr = plasma creatinine concentration = 8.
V = urine flow rate = 576 mL/24 hours = 24 mL/hour = 0.40 mL/min. Since the clearance values in the options are all in mL/m, the urine volume must be calculated in mL/min from the date given.

Ccr = 230 mg/dL x 0.4 / 8 mg/dL
= 11.5 mL/min

Question #: 2

A 50-year-old man comes to the physician because of a 1-week history of increased thirst and frequent urination. He was diagnosed with diabetes insipidus 1 year ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following free water clearance values is most likely in this patient?

A. -3
B. -2
C. -1
D. 0
✓E. +2

Rationale: Free water clearance is the volume of pure water that is lost in urine in a day. This can be a positive value, negative value or zero. A positive value indicates pure water loss from the body through the kidney. A negative value indicates pure water gain by the body by water conservation in the kidney. Zero indicates no net loss or gain of pure water and the urine osmolarity is same as that of plasma. In diabetes insipidus, lack of ADH causes loss of water in urine and therefore the free water clearance is a positive value. All other options in the question are either negative or zero. Positive value is also indicative of the production of hypo-osmotic urine.

Question #: 3

A 38-year-old man comes to the physician because of a 2-week history of increased thirst and frequent urination. He has no major illness. His vital signs are all within normal limits. Physical examination shows no abnormalities. Laboratory studies show a hypotonic urine and decreased serum antidiuretic hormone concentration. A water deprivation test is performed, and an antidiuretic hormone is administered. Following this administration, his urine osmolality markedly increased. Which of the following is the most likely diagnosis?

✓A. Central diabetes insipidus
B. Nephrogenic diabetes insipidus
C. Type 1 diabetes mellitus
D. Syndrome of inappropriate diuretic hormone (SIADH)
E. Primary polydipsia

Rationale: Decreased ADH points toward central DI. The ADH challenge test confirms this suspicion because urine osmolarity increases to more normal levels after ADH administration. Accordingly, central DI is most likely.
Nephrogenic DI does not have decreased ADH and with that condition urine osmolarity should not change after the ADH challenge, hence nephrogenic DI is unlikely. Type I DM could also result in increased thirst and micturition due to glucose secretion, but that is the only symptom consistent with that diagnosis. Decreased ADH and more dilute urine are not consistent with type I DM. SIADH should come with increased ADH levels, not decreased as described in the vignette. Polydipsia also should not affect ADH levels.

Question #: 4

A 31-year-old woman comes to the physician because of a 3-week history of increased thirst and urination. She says she just recovered from a respiratory infection that was treated with a 10-day course of antibiotics. He has no major illness. Her blood pressure is 114/74 mm Hg; other vital signs are within normal limits. Physical examination shows dry tongue and mouth, and poor skin turgor. Laboratory studies show increased serum concentration of sodium and osmolality, reduced urine osmolality, and absent glucose in urine. She is diagnosed with nephrogenic diabetes insipidus because of antibiotic treatment. Which of the following renal changes is most likely in this patient?

A. Increased GFR
B. Decreased serum concentration of ADH

✓C. Decreased urea concentration in the medullary interstitium
D. Diminished reabsorption of Na+ in PCT
E. Reduced reabsorption of glucose in PCT

Rationale: This patient has diabetes insipidus characterized by reduced response of the renal tubules to ADH. There is increased serum ADH concentration, but the hormone is unable to act on the kidney. This results in reduced water reabsorption in the collecting duct and reduced urea movement into the renal interstitial space in the inner medullary collecting duct. The reduced urea movement causes reduced medullary osmolality.

Question #: 5

A 23-year-old man comes to the physician because of a 2-day history of passing large volume of urine and increased thirst. Three weeks ago, he was involved in a road traffic accident and hospitalized for a week because of head injury. His vital signs are within normal limits. Laboratory studies show a low serum concentration of ADH. Which of the following renal alterations is most likely because of the ADH deficiency in this patient?

A. Increased glomerular filtration rate
B. High osmolarity of tubular fluid leaving the thick ascending limb
C. Reduced urea reabsorption in the proximal tubule
D. Elevated renal medullary osmotic gradient
✓E. Low osmolarity of tubular fluid at the end of the cortical collecting duct

Rationale: ADH effect in kidney is to increase water reabsorption. This is done by increasing permeability of the distal nephron for water and by increasing the renal medullary osmotic gradient. When ADH is absent, these effects are not there which results in a hypotonic urine excretion (option E).
Other options: GFR is not affected by ADH, and it may be low if water loss is heavy (option A). The osmolarity of the tubular fluid leaving thick ascending limb is always low in all conditions (option B). Urea reabsorption in PCT occurs independently of ADH (option C). The renal medullary gradient would not increase (option D).

Question #: 6

A 63-year-old woman is brought to the physician by her husband because of a 2-day history of forgetfulness and confusion. She has no other medical issues and is not on any regular medications. Her husband says that she has been drinking a lot of plain water for the past few days. Her blood pressure is 120/66 mm Hg, and pulse is 82/min. She is mildly disoriented and confused. Laboratory tests show that her blood glucose is normal, but serum sodium concentration is 128 mEq/L (Normal: 135-145 mEq/L). Which of the following best explains this patient's altered mental status?

A. Decreased intracellular fluid volume

B. Increased intracellular osmolarity
✓C. Decreased plasma osmolarity
D. Increased plasma osmolarity
E. Low blood pressure

Rationale: This patient is drinking a lot of plain water for a few days. This causes reduction in the sodium concentration (hyponatremia) in the blood. Sodium is the major electrolyte that contributes to the osmolarity of plasma and therefore, plasma osmolarity decreases (option C). Reduction in the osmolarity of serum makes water move into the cells and increase ICF volume also. Increase in ICF volume makes the cells increase in size (cells swell). This also occurs in the brain which results in altered mental status in this patient.

Question #: 7

A 35-year-old man is brought to the emergency department following a car crash that occurred an hour ago. His temperature is 35.1°C (95.2°F), pulse is 110/min, blood pressure is 90/60 mm Hg, and respirations are 22/min. Physical examination shows open fractures of his left radius and ulna. Laboratory studies show a hematocrit of 32% and a plasma volume of 3 Liters. Which of the following values best approximates the blood volume of this patient?

A. 3.3 L
B. 4.0 L
C. 4.3 L
✓D. 4.4 L
E. 5.4 L

Rationale: Hematocrit indicates the fraction of red blood cell volume. Hematocrit 32% means that the volume of red cells is 32 mL in a total blood volume of 100 mL. That means that the volume of plasma is the remaining fraction (100-32 = 68). Normally hematocrit is represented by either percentage (out of 100) or a fraction out of 1. If it is given in percentage, it must be reduced to a fraction (out of 1) and then applied in the formula below.
Hematocrit = 32% = 32/100 = 0.32.

Blood volume = Plasma Volume / 1 - Hematocrit

BV = PV / 1 - Hct
= 3 / 1 - 0.32 = 3/0.68 = 4.4 L

Question #: 8

A 31-year-old man participates in a study related to estimation of various body fluids. His vital signs are within normal limits. He is injected intravenously with a radioactive sodium solution. The data during the study are tabulated as below:
Amount of radioactive sodium injected =380 mmol
Amount of radioactive sodium lost in urine within an hour =20 mmol

Concentration of radioactive sodium in serum after one hour =0.02 mmol/L.
Assuming that the radioactive sodium is distributed in his body fluid during 1-hour equilibration time. Which of the following values best represents the extracellular fluid volume in this person?

A. 15 Liter
✓B. 18 Liter
C. 24 Liter
D. 30 Liter
E. 42 Liter

Rationale: Indicator dilution method is used to calculate ECF by using the formula,
V = Q/c.
The indicator used is radioactive sodium which gets distributed in the extracellular fluid compartment at equilibrium. Therefore, this indicator measures the extracellular fluid volume.
Apply the values given in the data as:
Q = 380-20 = 360 mmol
C = 0.02 mmol/L
ECF volume = 360/0.02 = 18000 mL = 18 Liter.

Question #: 9

A 28-year-old man comes to the physician for a routine examination 1 month after surgery to remove a kidney stone. His blood pressure is 120/76 mmHg. The physician receives the following laboratory data from his recent renal clearance study:
Hydrostatic pressure in glomerular capillaries =59 mm Hg
Hydrostatic pressure in Bowman's Space =18 mm Hg
Oncotic pressure of filtrate in Bowman's Space =1 mm Hg
Oncotic pressure of plasma in glomerular capillaries =32 mm Hg
Filtration coefficient (Kf) =11 mL/min/mm Hg
RPF =600 mL/min
Which of the following values best approximates this patient's filtration fraction?

A. 0.10
✓B. 0.18
C. 0.2
D. 0.25
E. 0.5

Rationale: First calculate GFR from the values given. Apply this value in the formula to find Filtration Fraction (FF).
FF = GFR/RPF

Net filtration pressure = (Forces favoring filtration) - (Forces opposing filtration)
= (59+1) - (32+18) = 10.
GFR = Kf x P = (11) x (10) = 110 ml/min

FF = GFR/RPF = 110/600 = 0.18

Question #: 10

A 16-year-old girl is brought to the emergency department by her parents because of a 3-day history of fever, shortness of breath, muscle pain, and difficulty walking. Her blood pressure is 160/120 mm Hg, and pulse is 84/min. An oral administration of an anti-hypertensive drug and a beta-blocker is not effective in reducing her blood pressure. Ultrasound imaging of kidneys shows decrease in the size of the kidneys and calcification of the renal arteries, causing bilateral renal artery stenosis. Urinalysis shows significantly reduced urine output because of a reduction in the glomerular filtration. A decrease in which of the following best explains the reduced GFR in this patient?

A. Afferent arteriolar resistance
B. Glomerular capillary filtration coefficient
C. Protein concentration in Bowman's space
D. Efferent arteriolar resistance
✓E. Glomerular capillary hydrostatic pressure

Rationale: Stenosis of the renal artery causes reduction in the blood flow through the glomerular capillaries. This results in reduction in the glomerular capillary hydrostatic pressure (option A) which causes reduction in the GFR. A decrease in afferent or efferent arteriolar resistance occurs if any factor selectively produces vasodilation in these vessels (options A and D). A decrease in filtration coefficient occurs due to changes in the surface area or permeability of the filtration barrier as in glomerular diseases (option B).

Question #: 11

A 25-year-old man is brought to the emergency department by a rescue team from a mountain expedition site. His blood pressure is 80/62 mm Hg, and pulse is 86/min. He is conscious but his lips, mouth and eyes are dried because of severe dehydration. His urine output has decreased significantly. There is no proteins lost in his urine. Serum studies show elevated angiotensin II concentration. This hormone causes vasoconstriction of the efferent arterioles. Which of the following changes of the Starling forces at the peritubular capillary is most likely in this patient?

A. Hydrostatic pressure increases
B. Hydrostatic pressure does not change
C. Oncotic pressure does not change
D. Oncotic pressure decreases
✓E. Oncotic pressure increases

Rationale: Patient with severe dehydration leads to activation of the Renin-Angiotensin-Aldosterone System (RAAS) leading to increased production of Angiotensin II. Angiotensin II causes constriction of the efferent arteriole, which leads to an increase in glomerular capillary hydrostatic pressure as well as a decrease in renal plasma flow. This leads to an increase in filtration fraction (increased fraction of fluid being filtered out of the total plasma flow). Increased filtration leads to more protein free fluid being filtered into the Bowman space and a higher concentration of protein remaining. Hence the peritubular capillary oncotic pressure increases which favors more fluid reabsorption in the PCT.

Question #: 12

A 25-year-old man is brought to the emergency department 1 hour after he was involved in a motor vehicle collision in which he sustained burn injuries. His temperature is 37°C (98.6°F), blood pressure is 98/62 mmHg, pulse is 124/min, and respirations are 20/min. Physical examination shows second, and third degree burns on his face, trunk, and arms. He is diagnosed with hypovolemic shock and is resuscitated with intravenous fluids. Four hours later, his urine output significantly decreases, and his blood pressure does not improve. Serum studies show electrolytes and albumin concentrations within normal range. Which of the following best explains a reduction in the urine output of this patient?

A. Increased glomerular capillary hydrostatic pressure
✓B. Decreased glomerular capillary hydrostatic pressure
C. Increased glomerular capillary oncotic pressure
D. Decreased glomerular capillary oncotic pressure
E. Increased glomerular filtration coefficient

Rationale: This patient is in hypovolemic shock and has low blood pressure. Glomerular filtration depends on the renal blood flow and the glomerular capillary hydrostatic pressure. His glomerular capillary hydrostatic pressure is low because of low blood pressure. This reduces GFR and thus reduces urine output.

Question #: 13

A 30-year-old man is brought to the emergency department by his wife because of a 2-hour history of chest pain and shortness of breath. An ECG shows flat T waves in multiple leads. Laboratory studies show a serum potassium concentration of 3 mmol/L (normal range 3.5 - 5 mmol/L). Imaging his kidney and urinary bladder shows cortical damage of the right kidney, leading to the poor reabsorption of potassium from the kidney tubules. No pathology is visible in the left kidney. Which of the following mechanisms best explains potassium reabsorption in the proximal tubules of this patient's left (normal) kidney?

✓A. Occurs predominantly by solvent drag via paracellular pathways
B. Is coupled to H^+ secretion
C. Accounts for less than 50% of K^+ reabsorption in the nephron
D. Is sensitive to circulating levels of ADH
E. Is coupled to the reabsorption of glucose and amino acids

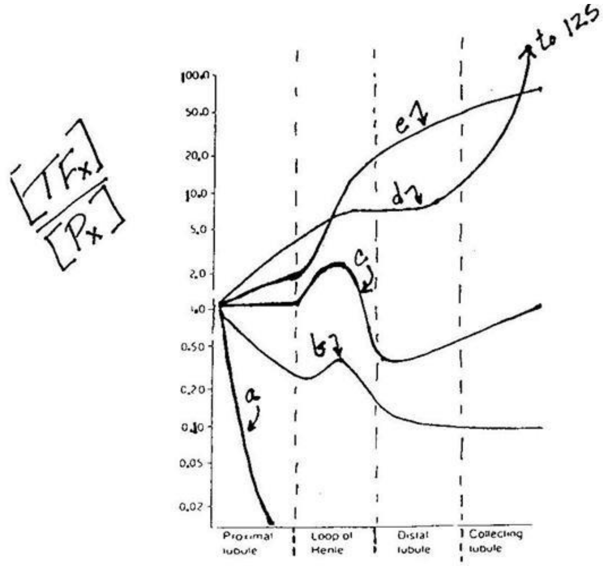
Rationale: The question is asking about the normal mechanism of reabsorption of potassium in the PCT. Normally, approximately 2/3 (66%) of filtered sodium, potassium, chloride, and water are reabsorbed in the PCT. This fraction of these solutes gets reabsorbed under all conditions and do not require any hormones. Potassium reabsorption in this part of the nephron is by solvent drag. The solvent drag is created when a large number of solutes move from the tubular lumen to the peritubular space in the PCT. The major solutes include sodium, chloride, glucose, bicarbonate, and amino acids. Water moves to the peritubular space by osmosis. Movement of fluid from this site to the peritubular capillaries takes place by Starling's forces. This fluid movement creates a solvent drag that makes these solutes and water move into blood in the peritubular capillaries.

Question #: 14

A 32-year-old woman comes to the physician because of a 1-week history of back pain and painful urination. She has a history of frequent urinary tract infections. An ultrasound of the kidneys and urinary bladder indicates cortical scars. Her tubular fluid/plasma analysis graph is attached. Which of the following best represents the relative concentration of urea along her kidney tubules?

A. A
B. B
C. C
D. D
✓E. E

Rationale: Urea reabsorption in the kidney is a passive process. Urea can freely move in and out of the tubules except in the inner medullary collecting ducts. In PCT and Loop of Henle, urea concertation in tubular fluid is slightly higher than that in plasma. However, in distal nephron, as larger proportion of water is reabsorbed than urea, the urea concentration in the tubular fluid increases 1.5 times, as noted and represented by curve E in the figure.
Other curves are:
Curve a indicates glucose; curve b indicates bicarbonate, curve c indicates sodium or chloride, and curve d represents PAH.
Attachment:



Renal Analysis_17.jpg

Question #: 15

A 46-year-old man comes to the physician because of a 1-month history of fatigue and shortness of breath. His blood pressure is 160/110 mm Hg. Physical examination shows generalized pitting edema. Results of laboratory studies of his blood and urine are shown:
Plasma creatinine: 1.5 mg/dL (normal: 0.6-1.2 mg/dL)
Glycosylated hemoglobin (HbA1c): 5.8% (normal: <6%)
Urine creatinine: 200 mg/dL or 2 mg/mL
24-hour urine volume: 1.44 L or 1 mL/min
Urine PAH: 250 mg/dL
Which of the following values best approximates his GFR?

A. 80 mL/min
B. 98 mL/min
C. 106 mL/min
D. 126 mL/min
✓E. 133 mL/min

Rationale: 133 mL/min is the correct answer. Creatinine clearance gives approximate value of GFR.
GFR calculation is done as follows:
$CCr = U_{Cr} \times V/P_{Cr}$

Creatinine clearance = GFR = (200 x 1)/1.5 = 133 mL/min.

Question #: 16

A 62-year-old man comes to the physician because of a 3-day history of headache, mild chest pain on short walks, difficulty breathing and occasional blurred vision. His blood pressure is 164/96 mmHg, and pulse is 68/min. His plasma electrolytes are within the normal ranges. He is prescribed a diuretic drug that inhibits sodium-chloride reabsorption in the distal convoluted tubules. Which of the following groups of diuretics is most likely prescribed to this patient?

A. Osmotic diuretic
B. Loop diuretic
C. Potassium sparing diuretic
✓D. Thiazide diuretic
E. Carbonic anhydrase inhibitor

Rationale: Thiazide diuretics inhibit Na-Cl symporters in the DCT. This group of diuretics is the first choice to decrease hypertension (HTN). Thiazide diuretics lower HTN by inhibiting NaCl in the early distal tubule, increasing sodium excretion and causing diuresis. This reduces the plasma volume, decreases the cardiac output, and thus decreases blood pressure.

Question #: 17

A 25-year-old man comes to the physician because of a 10-day history of breath odor and headaches. He says that he adopted a ketogenic diet for weight loss 4 weeks ago. He is 180 cm tall and weighs 130 kg; BMI is 40.1 kg/m^2. Physical examination shows a fruity odor to his breath. Laboratory studies show serum glucose concentration is 75 mg/dL (Normal: 70-100 mg/dL). Urinalysis is positive for ketone bodies. Which of the following best describes his acid-base balance on this diet?

A. The renal carbonic anhydrase system is less active
✓B. Secreted protons are buffered by NH_3 to form NH_4Cl in the collecting duct
C. Urine pH is increased, and HCO_3^- is lost in urine
D. Renal glutaminase activity is very low
E. Beta-intercalated cells secrete H^+ ions into tubular lumen and reabsorb bicarbonate

Rationale: On a low-carbohydrate ketogenic diet, patient is producing ketone bodies (non-volatile acids).
Ketone bodies are weak acids which give off H+, which are buffered by HCO3- (bicarbonate).
Increased production of ketone bodies (acids), results in increased activity of renal carbonic anhydrase especially in the alpha-intercalated cells --> Increased tubular H+ secretion.
Urine pH would be low on ketogenic diets due to increased H+ losses in urine as acid phosphate and ammonium chloride.
Renal glutaminase activity is increased to form ammonia which binds to H+ in the renal tubule to form ammonium chloride which is lost in urine.
Beta-intercalated cells secrete HCO3- into tubules and reabsorb H+. They are active in conditions of chronic alkalosis.

Question #: 18

An 85-year-old woman is brought to the physician by her daughter because of a 3-day history of chest discomfort and generalized malaise. Two weeks ago, she developed pedal edema, and her physician prescribed a loop diuretic. She has a history of hypertension, type 2 diabetes mellitus, dyslipidemia, and Parkinson's disease and reports compliance with her medications. Which of the following acid-base disturbances is most likely to be observed in this patient after initiation of the diuretic?

A. Metabolic acidosis with hyperkalemia
B. Increased anion gap metabolic acidosis
✓C. Metabolic alkalosis with hypokalemia
D. Respiratory alkalosis with renal compensation
E. Acute respiratory acidosis
F. Respiratory acidosis with renal compensation

Rationale: Furosemide (loop diuretic) – causes decreased tubular Na+ reabsorption, leading to increased renal fluid loss. This results in ECF volume contraction --> Activation of Renin angiotensin aldosterone (RAAS) system --> Aldosterone results in increased renal tubular H+ and K+ secretion --> Metabolic alkalosis with hypokalemia.
Thiazide diuretics also have similar effects.

Question #: 19

An 18-month-old boy is brought to the physician by his parents because of a 3-day history of severe cough, shortness of breath, and fever. His medical history is notable for recurrent respiratory infections and chronic diarrhea since 6 months of age, requiring multiple hospital admissions. Laboratory studies show very low B-cell counts. A serum protein electrophoresis is performed, confirming a primary immunodeficiency. Which of the following protein fractions is most likely decreased in this patient?

A. Albumin
B. Alpha-1 globulin
C. Alpha-2 globulin
D. Beta globulin
✓E. Gamma globulin

Rationale: The vignette describes a child with repeated infections and immunodeficiency. The child also has reduced B-lymphocyte count and low immunoglobulin levels in serum protein electrophoresis. The child may have hypogammaglobulinemia (low levels of immunoglobulins) or agammaglobulinemia (absent immunoglobulin levels).
Other causes of immunodeficiency are SCID (due to autosomal recessive ADA deficiency or X-linked recessive interleukin receptor defect), DiGeorge syndrome (thymic aplasia and T-cell deficiency), Purine nucleoside phosphorylase deficiency (T-cell deficiency).

Question #: 20

A 58-year-old man has been admitted to the hospital for evaluation of jaundice and abdominal swelling for the past month. He has a 15-year history of alcohol use disorder. Physical examination shows jaundice and ascites. Laboratory studies show an AST: ALT ratio of 3:1. Serum protein electrophoresis is performed. Which of the following abnormalities in the serum protein profile is most likely observed in this patient?

✓A. A globulin beta-gamma bridge
B. A highly elevated albumin peak
C. A single elevated peak of monoclonal immunoglobulin
D. An elevated alpha-1 antitrypsin peak
E. A sharply elevated peak of alpha2-globulins

Rationale: The vignette describes a patient with chronic liver failure due to alcohol abuse. A characteristic finding on serum protein electrophoresis is the presence of a beta-gamma bridge in the globulin region. This is due to the increased secretion of immunoglobulins (IgA) which migrate to this region on electrophoresis.
The albumin levels are usually low in patients with chronic liver disease as there is reduced synthesis of albumin due to poor liver function. Albumin is generally synthesized by the liver.
Alpha-1 antitrypsin peak is elevated as an acute phase response in many inflammatory disorders, but it is not elevated in chronic alcoholic cirrhosis.
Alpha-2 globulin levels are elevated in acute phase response in inflammatory disorders. Ceruloplasmin and alpha-2 macroglobulin are proteins that migrate to this region. Patients with nephrotic syndrome may have elevated alpha-2 peaks due to accumulation of alpha-2 macroglobulin (which has a high molecular weight).
A single monoclonal peak in the gamma-globulin region indicates multiple myeloma. This is a tumor of immunoglobulin producing tumor of plasma cells (mature B-cells).

Question #: 21

A 26-year-old man is admitted to the hospital for evaluation of a 2-week history of fever, chills, and generalized weakness. Physical examination shows pallor. Laboratory studies show hemoglobinuria. Peripheral blood smear shows intraerythrocytic ring forms consistent with Plasmodium species. To assess the severity of hemolysis, the concentration of a specific serum protein is measured. Which of the following best describes this serum protein?

A. Binds to free heme in plasma
✓B. Binds to free hemoglobin in plasma
C. Transports iron in plasma
D. Negative acute-phase reactant
E. Protein of the beta-globulin fraction

Rationale: LOW serum haptoglobin levels are very good laboratory indicators of hemolysis. When hemoglobin is released from RBCs as occurs in hemolytic anemias, haptoglobin binds to

free hemoglobin in plasma and prevents loss of protein (globin) in urine. The haptoglobin-hemoglobin complex is next taken up by the liver and as a result, serum haptoglobin levels fall during acute hemolysis.
Hemopexin binds to free heme during hemolysis and prevents loss of iron in urine during an episode of hemolysis.
Haptoglobin migrates to the alpha-2 globulin region along with alpha2-macroglobulin and ceruloplasmin. It is a positive acute phase reactant and levels of haptoglobin rise following inflammation.
Transferrin is the transport protein for iron.

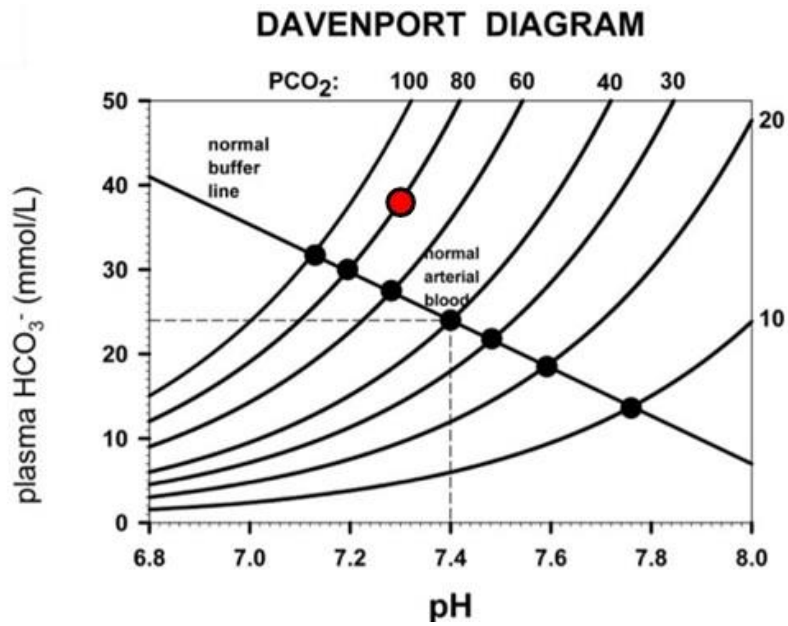
Question #: 22

A 58-year-old man is brought to the physician because of a 2-day history of worsening shortness of breath. His wife says that he has had a fever and a productive cough for 5 days. He has a 25-pack-year history of cigarette smoking. Physical examination shows cyanosis and respiratory distress. Auscultation of the lungs shows diffuse crepitations bilaterally. Arterial blood gas studies on room air are represented by the red dot on the Davenport diagram shown. Which of the following compensatory mechanisms is most likely active in this patient?

A. Decreased serum bicarbonate levels
B. Increased urinary bicarbonate excretion
C. Inhibition of renal carbonic anhydrase
✓D. Increased urinary ammonium excretion
E. Increased rate and depth of respiration

Rationale: The point depicted on the Davenport diagram shows low pH, PCO2- 80 isopleth and HCO3 =38 (Increased). This represents chronic respiratory acidosis.
Acute respiratory acidosis would be on the blood-buffer line. pH - low; PCO2 increased; HCO3 - normal
Mixed acidosis is characterized by low pH; HCO3 - low and PCO2 - increased
Metabolic alkalosis - Increased pH; HCO3 - Increased and PCO2 - Increased
Metabolic acidosis - Low pH; HCO3- low; PCO2 - low

Plot the other disorders on the Davenport diagram for practice.
In chronic respiratory acidosis, the renal system compensates by increasing the tubular H+ secretion and increasing the serum HCO3- levels. Renal carbonic anhydrase is stimulated by acidosis. The secreted H+ are buffered by phosphate and ammonia resulting in increased excretion of acid phosphate and ammonium in urine.
Attachment:



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Question #: 23

A 45-year-old man presents to the emergency department with a 2-week history of fatigue and muscle cramps. Laboratory studies show metabolic alkalosis, increased renal tubular H^+ ion secretion, and a persistently low urine pH. Which of the following conditions is most likely responsible for these findings?

A. Decreased arterial PCO2
B. Adrenal cortical insufficiency
C. Administration of carbonic anhydrase inhibitor
D. Diabetic ketoacidosis
✓E. Hypokalemia

Rationale: Renal tubular H+ secretion is dependent on carbonic anhydrase activity.
Increased tubular H+ secretion is observed in:
1. Increased arterial PCO2 (respiratory acidosis)
2. Decreased intracellular pH (acidosis; In metabolic acidosis, renal H+ secretion increases only if the kidneys are functional)
3. Excessive aldosterone levels. Note that aldosterone stimulates the H+ pump in the renal tubules and results in increased H+ losses in urine and metabolic alkalosis in hyperaldosteronism.
4. Hypokalemia (Correct answer) stimulates the H+ secretion in the renal tubules. Hypokalemia is always associated with alkalosis due to renal tubular losses of H+.

H+ secretion in renal tubules decreases in:
1. Administration of acetazolamide which inhibits renal carbonic anhydrase.
2. Aldosterone deficiency (Addison disease)
3. Alkalosis (increased arterial or intracellular pH) - inhibits the renal carbonic anhydrase system

4. Decreased arterial PCO2 (respiratory alkalosis) inhibits carbonic anhydrase system and decreases H+ secretion in urine; Urinary pH increases and there is loss of HCO3 in chronic respiratory alkalosis.

Question #: 24

A 4-year-old boy is brought to the emergency department by his mother because of a 1-hour history of abdominal pain and vomiting. She says that she found an empty bottle of aspirin tablets (acetyl salicylic acid). His respirations are 32/min. Pulse oximetry on room air shows oxygen saturation of 95%. Arterial blood gas analysis on admission show:

- pH =7.5 (Normal: 7.35-7.45)
- PCO_2 = 30 mm Hg (Normal: 33-45 mm Hg)
- Bicarbonate (HCO_3^-) =23 mEq/L (Normal: 22-28 mEq/L)

Which of the following acid-base disorders is most likely in this patient?

A. Chronic respiratory acidosis due to poor tissue oxygenation
✓B. Acute respiratory alkalosis due to hyperventilation
C. Metabolic acidosis due to loss of bicarbonate
D. Chronic respiratory alkalosis with renal compensation
E. Acute respiratory acidosis due to hypoventilation
F. Mixed alkalosis

Rationale: Aspirin toxicity can result in respiratory alkalosis. Aspirin stimulates the respiratory center and increases the respiratory rate. Hyperventilation results in a fall in PCO2 and a rise in pH. In acute respiratory alkalosis, the serum Bicarbonate is normal. In chronic respiratory alkalosis, HCO3 levels decrease due to renal compensation.
Patients with aspirin poisoning may also be present with metabolic acidosis. Aspirin (acetyl salicylate) poisoning can also cause metabolic acidosis. If the ABG shows low pH; Low HCO3- and low PCO2 it indicates metabolic acidosis

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Question #: 25

A 20-year-old man is brought to the emergency department after a motor vehicle collision. His blood pressure is 70/60 mm Hg, and pulse is 100/min. Physical examination shows a rapid and weak pulse, multiple long-bone fractures, and extensive bruising of the extremities. Initial laboratory and arterial blood gas studies on room air are shown in the table. Which of the following is the most likely cause of this acid-base disturbance?

✓A. Lactic acidosis
B. Respiratory distress syndrome

C. Nasogastric suction
D. High altitude sickness
E. Pyloric stenosis
F. Diarrhea

Rationale: The ABG analysis shows low pH; Low HCO3 and low PCO2. This indicates metabolic acidosis.
The anion gap is calculated: Na - (Cl- +HCO3-) = 142 - 118 = 24 mEq/L (Normal Anion gap is 8-12 mEq/L)
The patient has high anion gap metabolic acidosis (Increased anion gap metabolic acidosis).
Of the listed disorders, Diarrhea causes normal anion gap acidosis.
In Diabetic ketoacidosis, low levels of insulin results in increased production of ketone bodies.
Ketone bodies act as 'unmeasured anions' and result in increased anion gap.
Nasogastric suction and pyloric stenosis can result in metabolic alkalosis.
High altitude sickness results in respiratory alkalosis.
Respiratory distress syndrome causes respiratory acidosis (due to poor gas exchange and retention of CO2)

Attachment:

Analyte	Patient	Reference range
Na^+	142	136-145 mEq/L
Cl^-	98	95-105 mEq/L
$PaCO_2$	25	33-45 mmHg
HCO_3^-	10	22-28 mmol/L
pH	7.22	7.35 – 7.45

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Question #: 26

A medical student reviews arterial blood gas studies for a patient:

- pH: 7.50 (normal: 7.35–7.45)
- $PaCO_2$: 48 mm Hg (normal: 33-45 mm Hg)
- Bicarbonate (HCO_3^-): 32 mmol/L (normal: 22–28 mmol/L)

Which of the following clinical conditions best explains these findings?

✓A. A 3-week-old boy with a 4-day history of projectile vomiting
B. A 45-year-old man who has been on treatment for chronic obstructive lung disease presenting with fever, cough, and chest pain
C. A 20-year-old woman who has been camping at a high altitude for the past two days and has been hyperventilating

D. A 20-year-old man with type 1 diabetes mellitus with ketoacidosis
E. A 60-year-old man who has an airway obstruction due to a tumor in the right bronchus
F. A 2-year-old boy with severe diarrhea for three days

Rationale: The arterial blood gas findings indicate metabolic alkalosis due to severe vomiting. Vomiting causes loss of stomach HCl which results in elevated serum HCO3, increased pH, and increased PaCO2 (due to compensatory hypoventilation)
Diabetic ketoacidosis results in high anion gap metabolic acidosis. Ketone bodies are weak acids and are the unmeasured anions contributing to the increased anion gap.
COPD with infection may result in respiratory acidosis.
High altitude sickness results in respiratory alkalosis.
Airway obstruction may result in acute respiratory acidosis.
Diarrhea may result in normal anion gap metabolic acidosis due to loss of bicarbonate rich intestinal secretions.

Question #: 27

A 50-year-old man undergoes elective surgery and receives preanesthetic medication containing morphine. Two hours later, he develops pinpoint pupils, shallow respirations, and confusion. Respirations are 6/min. Physical examination shows slow and shallow breathing. Which of the following sets of arterial blood gas (ABG) findings is most likely observed in this patient?

A. I
B. II
✓C. III
D. IV
E. V

Rationale: Morphine toxicity may result in respiratory center depression. This results in reduced rate of ventilation and accumulation of CO2 resulting in an increase in PCO2. Accumulation of CO2 results in a fall in pH resulting in acute respiratory acidosis. The HCO3 levels are within normal limits as the renal system needs time to compensate for the respiratory acidosis.

Attachment:

	pH	PCO_2	HCO_3^-
I	Normal	Increased	Decreased
II	Increased	Decreased	Increased
III	Decreased	Increased	Normal
IV	Increased	Increased	Increased
V	Decreased	Decreased	Normal

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Question #: 28

A 16-month-old boy is brought to the physician by his mother because of a 3-day history of fever and severe diarrhea. His pulse is rapid and weak, blood pressure is low, and respiratory rate is increased. Physical examination shows sunken eyes and decreased skin turgor. Arterial blood gas studies on room air are obtained. Which of the following sets of findings is most likely to be observed in this patient?

A. I
✓B. II
C. III
D. IV
E. V
F. VI

Rationale: In patients with diarrhea, there is a loss of HCO3 in the intestinal secretions. As a result, serum HCO3 decreases which results in a fall in the pH - metabolic acidosis. Acidosis stimulates the respiratory center and increases the rate and depth of respiration (hyperventilation) and there is a fall in PCO2. The respiratory system compensates immediately (minutes to hours)and hence PCO2 is low on admission. The anion gap is normal as there is no accumulation of weak acids like lactate or ketone bodies.

pH-↓; PCO2- ↓; HCO3--↓; Anion gap-increased - These lab findings are typically found in lactic acidosis, ketoacidosis, or renal failure.
Anion gap is used in metabolic acidosis to determine the cause of the disorder and to identify whether there is accumulation of an 'unmeasured anion'.
pH-↑; PCO2-↑; HCO3--↑ - These findings are typical of metabolic alkalosis
pH-↓; PCO2-↑; HCO3--↑ - Chronic respiratory acidosis
pH-↑; PCO2-↓; HCO3--↓ - Chronic respiratory alkalosis

Attachment:

	pH	PCO_2	HCO_3^-	Anion gap
I	Increased	Decreased	Decreased	Normal
II	Decreased	Decreased	Decreased	Normal
III	Decreased	Increased	Increased	Increased
IV	Decreased	Decreased	Decreased	Increased
V	Decreased	Normal	Decreased	Normal
VI	Increased	Increased	Increased	Increased

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Question #: 29

A 27-year-old man comes to the physician because of a 2-day history of headache and nausea. He recently moved from New Jersey (sea level) to Colorado Springs, which is at 4300 m (14,000 ft) above sea level. Arterial blood gas studies on room air show:

- pH: 7.49 (Normal: 7.35–7.45)
- PCO_2: 25 mm Hg (Normal: 33–45 mm Hg)
- Bicarbonate (HCO_3^-): 24 mmol/L (Normal: 22–28 mmol/L)

Which of the following changes is most likely observed after one week in this patient?

✓A. An increase in urinary bicarbonate
B. Inhibition of erythropoietin secretion
C. Stimulation of renal carbonic anhydrase
D. An increase in urinary acid phosphate excretion
E. An increase in renal tubular hydrogen ion secretion
F. Decreased erythrocyte 2,3-bisphosphoglycerate levels

Rationale: The patient is in acute respiratory alkalosis. Hypoxia at high altitude stimulates the respiratory center and increases the rate and depth of respiration (hyperventilation)
Note the low $PaCO_2$ resulting in an increase in pH. The HCO_3 levels are within normal limits. In the chronic state, the renal compensation will be active and will decrease the serum bicarbonate. There is decreased H^+ secretion in urine, decreased acid phosphate and ammonium losses in urine, leading to increased urine pH. Urinary HCO_3 excretion is increased. The respiratory system is responsible for primary acid base disturbance and cannot assist in compensation.
Hypoxia stimulates 2,3-BPG formation in RBCs which facilitates oxygen unloading from hemoglobin.
Hypoxia also stimulates renal erythropoietin which favors RBC formation in the bone marrow.

Question #: 30

A 13-year-old girl is brought to the emergency department by her parents because of a 1-day history of worsening nausea and abdominal pain. She has a 3-week history of increased thirst and urination. Laboratory studies on admission show:

- Serum glucose: 550 mg/dL (Normal: 70-100 mg/dL)
- Urinalysis: positive for glucose and ketone bodies

Which of the following arterial blood gas (ABG) results is most likely to be observed in this patient?

A. I
B. II
✓C. III
D. IV
E. V
F. VI

Rationale: The vignette describes a patient with diabetic ketoacidosis. Diabetic ketoacidosis is characterized by metabolic acidosis (low pH, low HCO3, and low PCO2).
The serum bicarbonate is reduced as it is lost by buffering the ketone bodies which are weak acids. The loss of bicarbonate by buffering results in the lowering of the arterial pH. Acidosis stimulates the respiratory center and increases the rate of respiration which results in a fall in the PCO2 (compensation). The respiratory system compensates within minutes, and hyperventilation is observed on admission. The anion gap is increased due to the presence of ketone bodies which are the unmeasured anions. The same findings are present in a patient with lactic acidosis.
Normal anion gap metabolic acidosis is seen in patients with diarrhea (IV)
Attachment:

	pH	PCO_2	HCO_3^-	Anion Gap
I	Decreased	Normal	Decreased	Increased
II	Decreased	Decreased	Increased	Normal
III	Decreased	Decreased	Decreased	Increased
IV	Decreased	Decreased	Decreased	Normal
V	Decreased	Increased	Normal	Increased
VI	Increased	Increased	Increased	Normal

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